Chapter 5: Arrays and Strings

Arrays and strings are fundamental concepts in C programming, offering powerful ways to store and manipulate collections of data. This chapter explores the basics of arrays, multidimensional arrays, and the handling of strings, including essential functions provided by the C Standard Library.

5.1 Introduction to Arrays

What is an Array?

An **array** is a collection of variables of the same data type stored in contiguous memory locations. Arrays allow you to manage large amounts of data efficiently by grouping related values under a single variable name. Each element in an array can be accessed using an index.

Declaring Arrays

To declare an array, you specify the data type and the size (number of elements) of the array.

Syntax:

data_type array_name[size];